v5 Layerwise Self-Distillation — Campaign Final Report

Gemma-4-31B, Machado + Quijote ch.1-4 — closed 2026-08-19; probe program 2026-08-20/22 — report 2026-08-23

HEADLINE (null-first): the objective optimizes but does not store content. Scaffold-stripped content recall moved ~0.12 -> ~0.13 vs a teacher at 0.99 across ~40 arms, while 50/60 per-layer losses fell monotonically. Per-position hidden proximity (delta_cosine and every tested variant) is satisfiable by register/geometry alignment without storing the token sequence.

THE LAWS ESTABLISHED

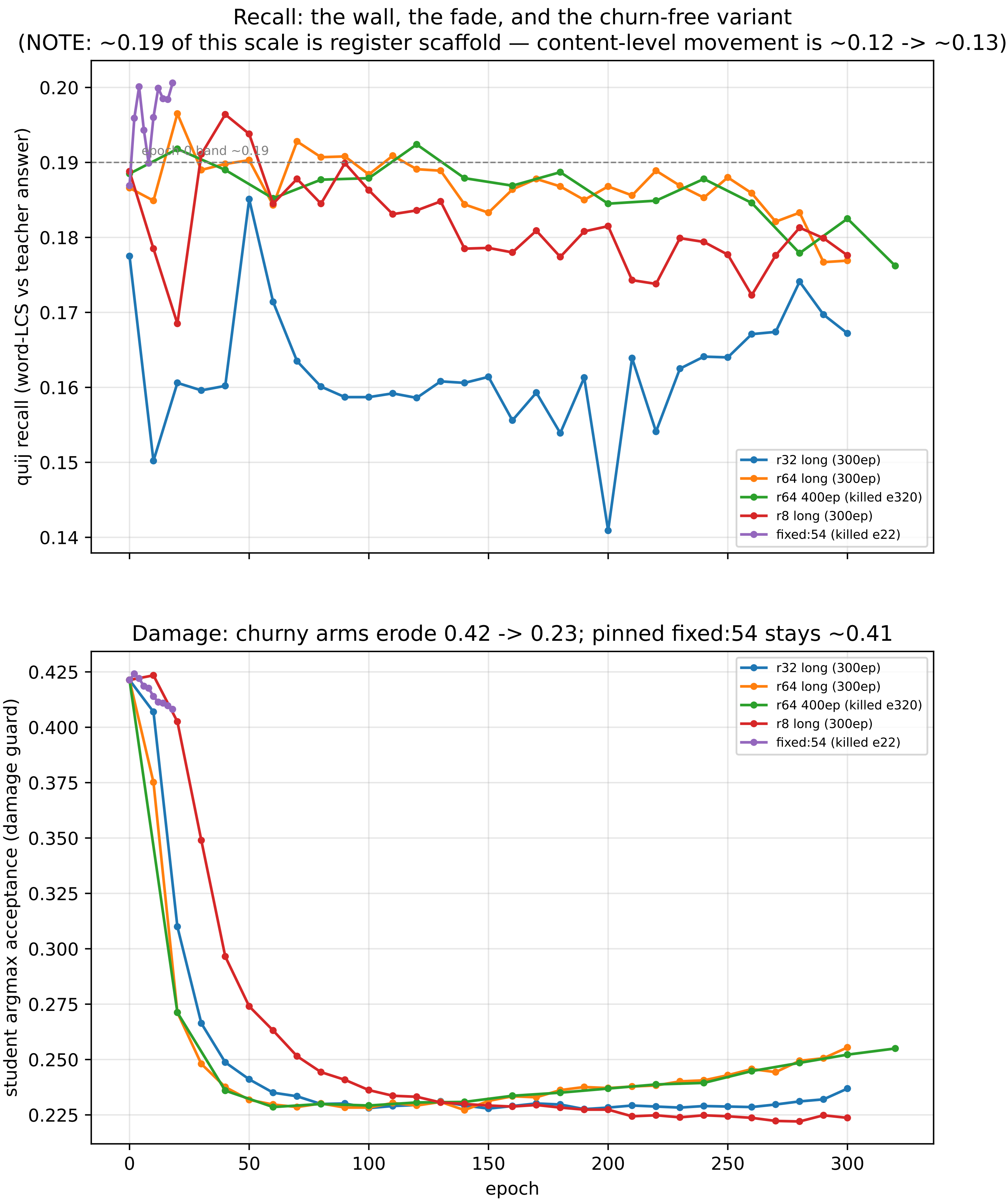
1. Content null: full-LCS recall carries ~0.19 of assistant-register scaffold shared with the teacher's answers; screen "gains" were scaffold +/- <=0.01 content (q4 content went DOWN in the best screen).
2. Rank-independent wall (~0.19): r8 = r32 = r64 = r128.
3. Peak-and-fade at every horizon (40/300/400 ep); churny runs end BELOW their own epoch-0.
4. Damage = churn: pinned fixed:54 reaches the same recall band with ZERO argmax/ARC erosion; per-step topk re-ranking pays argmax 0.42 -> 0.23.
5. Instabilities: increment-target (tinc) losses destroy at any gate width; dense isotropic losses destroy; DoRA null; norms arm contaminated (frozen-teacher bug, repaired: frozen_base()).
6. Write geography: gate collapse at L54 = one layer after Gemma's 4th-strongest retrieval global (L53).

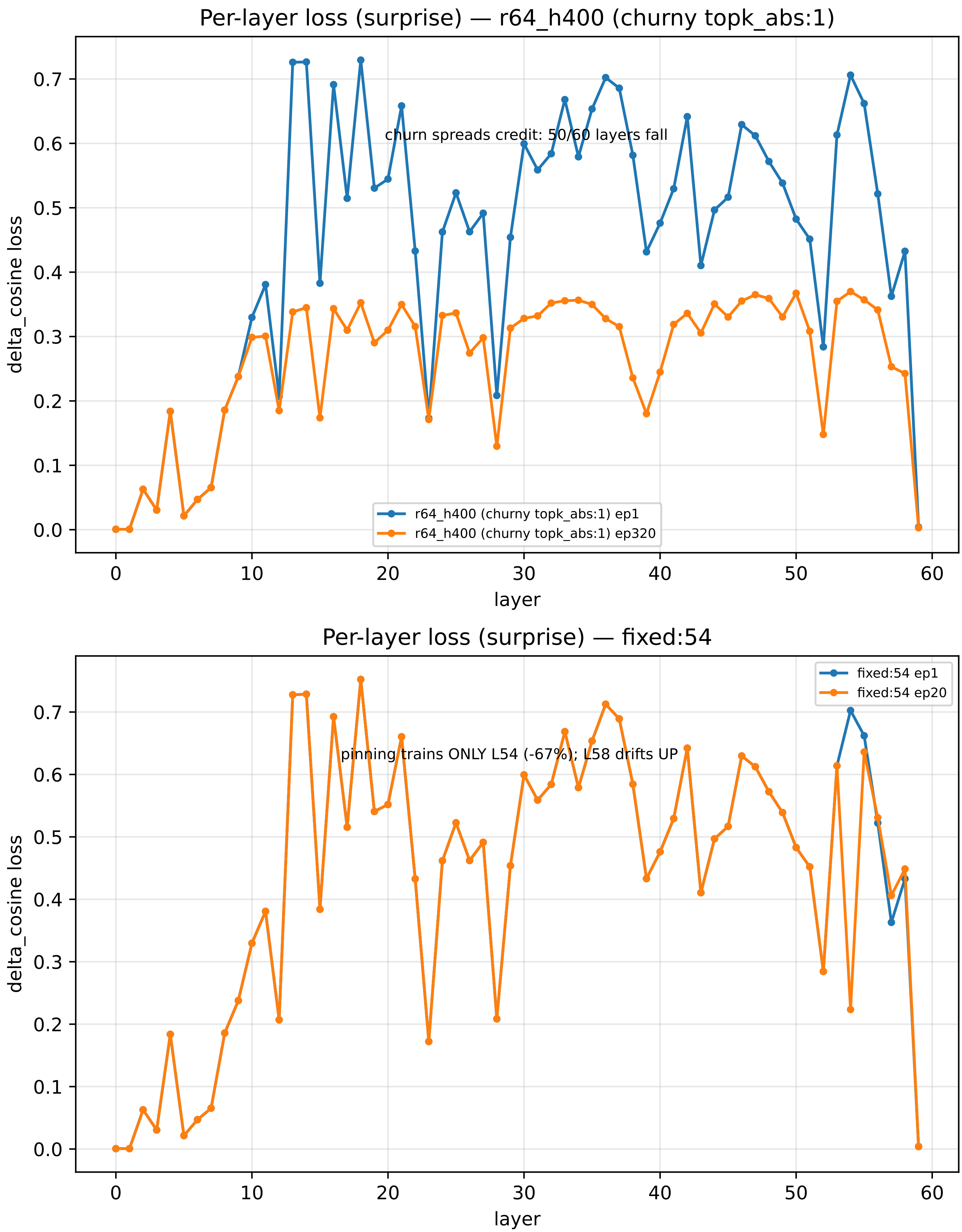
MECHANISM PROBES (CPU, Aug 20-22)

- Teacher attention: Gemma retrieves the passage via its 10 GLOBAL layers, deep-weighted (top: 47, 23, 41, 53). Qwen3.6 (GatedDeltaNet/softmax 3:1 hybrid): softmax layers 51, 55, 15, 59 lead ("early-stack" reading was an indexing artifact, retracted).
- Layer-censorship (retrieval-only mask, after fixing an encoding confound the owner caught): Gemma's 8 top globals are SUFFICIENT (CE 0.028 vs baseline 0.003) and NECESSARY (blocking them: CE 3.66) — a compact, causally verified retrieval locus.
- Qwen recurrent-only cell: with ALL softmax retrieval blocked, the DeltaNet state alone keeps 65% verbatim token accuracy (CE 1.58).
- Random-lesion seed: 8-layer subsets with <=1 global are near-harmless — redundancy lives in the globals.

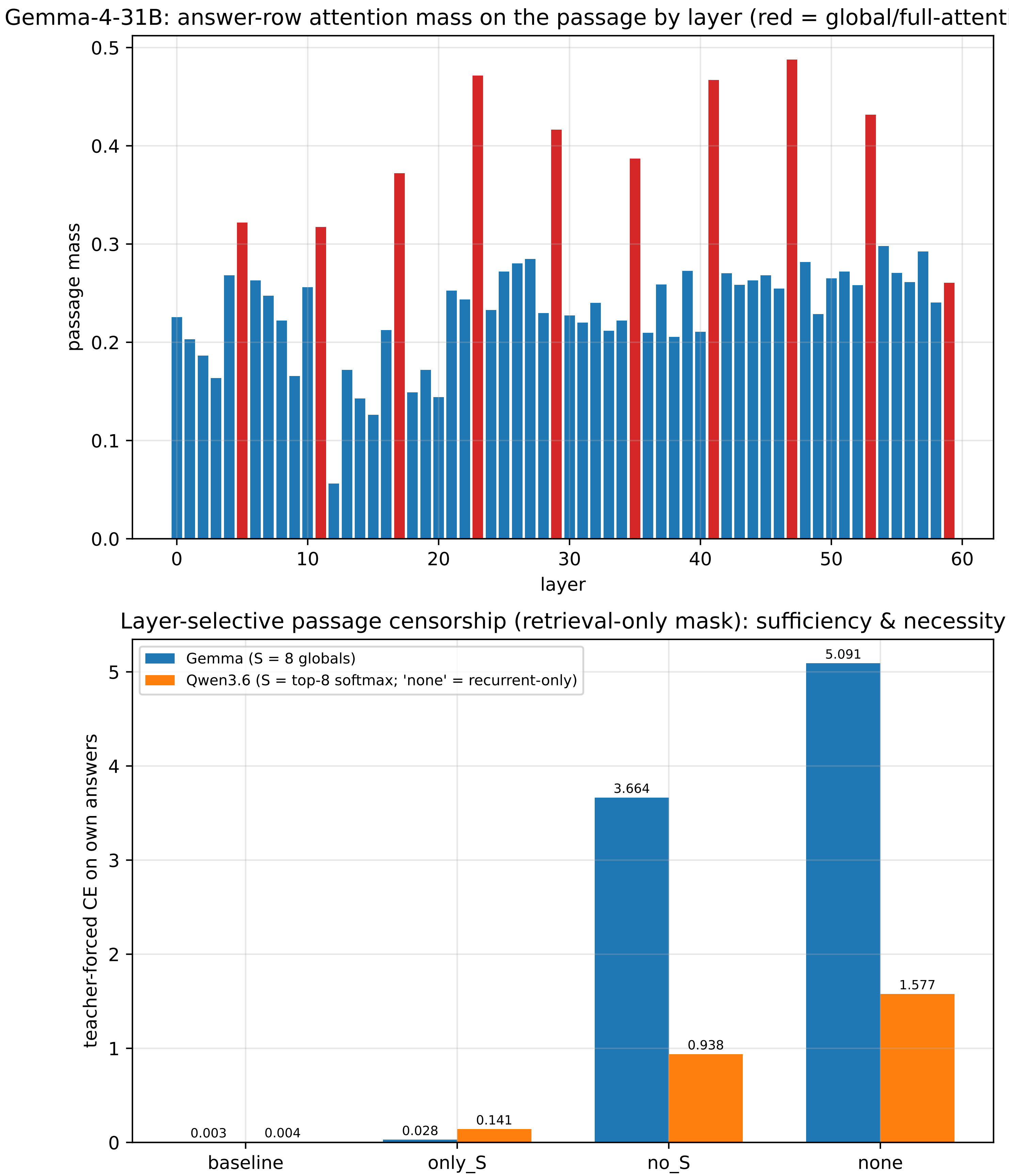
LICENSED SUCCESSOR (design E, owner): partial teacher censorship — a teacher seeing the passage ONLY at the 8 globals is behaviorally intact, making every non-global layer's target exactly reachable for a censored student and concentrating the storage burden at the globals. Opening v5w3 is an owner decision.

PENDING GPU ADDENDA (queued, cluster occupied): warm-start fade test (churn-vs-intrinsic), Qwen3.6 recall control (does the null generalize?), HF-side ceiling calibration, ALIA-40b answers artifact.


Recall: the wall, the fade, and the churn-free variant
(NOTE: ~0.19 of this scale is register scaffold — content-level movement is ~0.12 -> ~0.13)
quij recall (word-LCS vs teacher answer)
epoch 0 band ~0.19
0.20
0.19
0.18
0.17
0.16
0.15
0.14
r32 long (300ep)
r64 long (300ep)
r64 400ep (killed e320)
r8 long (300ep)
fixed:54 (killed e22)
Damage: churny arms erode 0.42 -> 0.23; pinned fixed:54 stays ~0.41
student argmax acceptance (damage guard)
0.425
0.400
0.375
0.350
0.325
0.300
0.275
0.250
0.225
r32 long (300ep)
r64 long (300ep)
r64 400ep (killed e320)
r8 long (300ep)
fixed:54 (killed e22)
0
50
100
150
200
250
300
epoch



Per-layer loss (surprise) — r64_h400 (churny topk_abs:1)
churn spreads credit: 50/60 layers fall
r64_h400 (churny topk_abs:1) ep1
r64_h400 (churny topk_abs:1) ep320
delta_cosine loss
layer
Per-layer loss (surprise) — fixed:54
pinning trains ONLY L54 (-67%); L58 drifts UP
fixed:54 ep1
fixed:54 ep20
delta_cosine loss
layer


Gemma-4-31B: answer-row attention mass on the passage by layer (red = global/full-attention layers
passage mass
0.0
0.1
0.2
0.3
0.4
0.5
0
10
20
30
40
50
60
layer
Layer-selective passage censorship (retrieval-only mask): sufficiency & necessity
Gemma (S = 8 globals)
Qwen3.6 (S = top-8 softmax; 'none' = recurrent-only)
teacher-forced CE on own answers
0
1
2
3
4
5
0.003
0.004
0.028
0.141
3.664
0.938
5.091
1.577
baseline
only_S
no_S
none